

# Refat Pasha

Software Engineer · .NET / Full-Stack

+880 1911827231 refatpasha567@gmail.com refat-pasha.github.io github.com/refat-pasha Dhaka, Bangladesh

Computer Science & Engineering graduate (CGPA **3.93/4.00**) building production-grade **ASP.NET Core** systems with **Angular** front ends. Shipped a 43-table university ERP with **389 passing tests** at **58.6% coverage** across 17 modules and 7 roles. Strongest in clean architecture, concurrency-safe data handling, and security-conscious design.

## SKILLS

**Backend** — C#, ASP.NET Core (MVC / Web API / SignalR), .NET 10, EF Core, LINQ**Frontend** — Angular 22, TypeScript, Tailwind CSS, Razor, Bootstrap**Database** — SQL Server, SQLite, EF Migrations, query optimization**Architecture** — Clean Architecture, SOLID**Auth & Security** — ASP.NET Identity, JWT, policy authorization**Testing & DevOps** — xUnit, integration tests, Git, Docker, Swagger**Also** — Flutter, Dart, Firebase, GetX, Python, Flask, PHP

## PROJECTS

### CampusConnect — University ERP & Student Portal FLAGSHIP

ASP.NET Core MVC · Web API · EF Core · SQL Server · SignalR · xUnit | [github.com/refat-pasha/CampusConnect](https://github.com/refat-pasha/CampusConnect)

- Production ERP: **17 modules, 43 tables, 7 roles** with 14 authorization policies.
- Rule engines (registration, grading, fees, scheduling, exams) written as **pure static classes** — unit-testable with no database or clock.
- REST API with a unified response envelope, meaningful status codes, and capped pagination.
- Hardened uploads against **stored XSS**, path traversal, and extension spoofing via magic-number validation.
- Dockerized with Compose; soft deletes, audit interceptors, live SignalR notifications.

### Flash Sale Movie Ticket Engine

ASP.NET Core 10 · Angular 22 · SignalR · SQL Server · xUnit | [github.com/refat-pasha/flashSaleMovieTicket](https://github.com/refat-pasha/flashSaleMovieTicket)

- Cinema ticketing built for **contention at scale** — thousands of buyers racing for the same seats.
- Killed double-booking with SQL `rowversion` optimistic concurrency → clean **409s** instead of silent overwrites.
- Virtual waiting room: FIFO queue + `BackgroundService` admitting batches every 10s over SignalR.
- Queue fairness enforced by a **server-validated JWT claim**, not just a client-side guard.
- Multi-seat basket holds in one transaction with partial-success handling; three sweeper workers reclaim abandoned inventory.

### TodoMaster — Task Management Web App

ASP.NET Core MVC · EF Core 10 · SQLite · Razor | [github.com/refat-pasha/ToDoMaster](https://github.com/refat-pasha/ToDoMaster)

- Full CRUD app with priority, due dates, server-side validation, and anti-forgery tokens on all POST actions.
- Structured Controllers → Services → DbContext → SQLite with async EF Core; documented path to Web API + Angular.

## EXPERIENCE

### Software Engineering Projects — Full-Stack .NET Developer

2026 – Present

- Built **3 production-grade .NET applications** end to end — architecture, schema, API, security, tests, deployment.
- Applied TDD and clean architecture throughout; authored architecture, security-checklist, and roadmap documentation.

### 11th Lesson — Learning Management System Team Developer

2023 – 2026

Flutter · Firebase (Auth/Firestore/Storage) · GetX · Flask · SQLAlchemy · PHP · MySQL | [github.com/refat-pasha/eeleventh\\_lesson\\_app\\_final](https://github.com/refat-pasha/eeleventh_lesson_app_final)

- Built a **3-year team project** from a static HTML site through Flask, PHP, and MySQL backends into a cross-platform **Flutter LMS**; owned the backend, database, mobile app, and UI/UX.
- Shipped Firebase Auth with student/teacher roles, quiz engine, assignment submission, study groups, offline library, and XP/streak gamification using GetX for state, routing, and DI.

### Machine Learning Research — Student Researcher

2024 – Present

- Built and evaluated classification/regression/clustering models in Python, scikit-learn, and TensorFlow; EDA and feature engineering on healthcare data.

## EDUCATION

### B.Sc. Computer Science & Engineering — Daffodil International University

CGPA 3.93 / 4.00 · 2026

Certifications: 365 Data Science (Data Scientist Track, ML in Python, TensorFlow 2, Pandas) · DataCamp (Python, SQL — Intro &amp; Intermediate)

Bengali native · English fluent